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Lecture 4: Experiments and covariate adjustment

Molly Offer-Westort

Department of Political Science,
University of Chicago

Notation: recall

- Dependent variable: (Y), the thing we want to study variation in; also "outcome", "response".
- Independent variable: (W) in causal settings, the thing that we manipulate. Also "treatment."
- In non-causal settings, something that may have some natural variation in the population associated with the dependent variable (X).

Potential outcomes: notation

W is binary.

$$Y_i^{obs} = \begin{cases} Y_i(0) & W_i = 0 \\ Y_i(1) & W_i = 1 \end{cases}$$

Estimand of interest: ATE

- Individual treatment effect

$$\tau_i = Y_i(1) - Y_i(0)$$

- Average treatment effect

$$\tau = \mathrm{E}\left[\tau_i\right] = \mathrm{E}\left[Y_i(1) - Y_i(0)\right]$$

- We're interested in the *average* difference in potential outcomes had we applied treatment W, vs. if we had not.

Difference-in-means

Proposed estimator for $\mathrm{E}[\tau_i]$: compare people who received treatment 1 to people who received treatment 0.

This is the difference in means estimator:

$$\begin{aligned}\hat{\tau}_{DM} &= \frac{\sum_i^n Y_i^{obs} W_i}{\sum_i^n W_i} - \frac{\sum_i^n Y_i^{obs}(1 - W_i)}{\sum_i^n (1 - W_i)} \\ &= \bar{Y}_t^{obs} - \bar{Y}_c^{obs}\end{aligned}$$

Regression version of this

$$Y_i^{obs} = \alpha + \tau \cdot W_i + \varepsilon_i$$

ε_i is an unobserved error term.
The "ordinary least squares" (OLS) regression solution is

$$\hat{\tau}_{ols} = \frac{\sum_{i=1}^{n}(W_i - \bar{W}) \cdot (Y_i^{obs} - \bar{Y}^{obs})}{\sum_{i=1}^{n}(W_i - \bar{W})^2} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs}$$
$$\hat{\alpha}_{ols} = \bar{Y}^{obs} - \hat{\tau}_{ols} \cdot \bar{W}$$

This estimate is equivalent to the difference-in-means estimate (check this).

Inference

- "randomization implies that the average residuals for treated and control units are zero"
- $\mathrm{E}\left[\varepsilon_i | W_i = 0\right] = 0$ and $\mathrm{E}\left[\varepsilon_i | W_i = 1\right] = 0$
- But random assignment does not imply errors are independent of W_i (what does this mean?)

$\rightarrow$ estimate with "robust" standard errors

Estimation

Using the Duflo et al. (2012) data,

$$\hat{Y}_i^{obs} = \underset{\substack{(0.0356) \\ [0.0311]}}{0.5805} + \underset{(0.0308)}{0.2145 \cdot W_i},$$

where Y is educational achievement, W is teacher presence.

Why adjust for covariates?

- improve precision (when covariates correlate with the outcome)
- assess heterogeneity
- account for imbalance in randomization

Balance checks

Olken (2010) Table 2

TABLE 2. Summary Statistics

	(1) Mean in Meeting Group	(2) Difference between Plebiscite and Meeting Group	(3) *p* Value	(4) Num Obs
Village characteristics				
Village population (1,000 inhabitants)	2.401	−0.295	0.625	49
	[2.726]	(0.598)		
Agricultural wage (1,000 Rupiah)	21.023	−1.061	0.466	43
	[5.892]	(1.443)		
Percent village roads that are asphalt	0.305	−0.042	0.507	49
	[0.269]	(0.062)		
Number of hamlets per village	4.813	−0.633	0.142	49
	[1.839]	(0.423)		
Number of churches and mosques per village	2.438	−0.220	0.698	49
	[1.933]	(0.563)		
Distance to subdistrict capital (km)	5.766	3.548	0.109	49
	[6.509]	(2.173)		
Village ethnic fragmentation	0.268	−0.075	0.190	49
	[0.250]	(0.056)		
Village religious fragmentation	0.106	0.011	0.827	49
	[0.137]	(0.051)		
Survey respondent characteristics				
Survey respondent predicted log per capita expenditure	11.505	0.034	0.602	224
	[0.279]	(0.066)		
Survey respondent years education	8.925	−0.519	0.404	244
	[3.088]	(0.616)		
Survey respondent is female	0.431	0.025	0.292	245
	[0.497]	(0.023)		
Survey respondent age	41.700	1.896	0.271	245
	[12.021]	(1.701)		
Survey respondent is farmer	0.594	−0.052	0.541	245
	[0.493]	(0.084)		

	(1) Mean in Meeting Group	(2) Difference between Plebiscite and Meeting Group	(3) *p* Value	(4) Num Obs
Village government characteristics				
Village head age	45.935	2.368	0.443	47
	[8.370]	(3.059)		
Village head years of education	11.645	−1.409	0.081*	47
	[2.026]	(0.788)		
Number of village head candidates in last village head election	2.207	0.304	0.432	44
	[1.013]	(0.383)		
More than one candidate in last village head election	0.724	0.089	0.449	44
	[0.455]	(0.116)		
Share of population that voted in last village head election	0.888	−0.004	0.910	43
	[0.100]	(0.031)		
Village head's margin of victory in last election (if challenger)	0.263	−0.011	0.870	33
	[0.262]	(0.069)		
Number of village government executive branch members	8.516	−0.616	0.386	47
	[2.850]	(0.703)		
Share of hamlets represented in village executive branch	0.853	0.043	0.442	47
	[0.240]	(0.056)		
Number of people in village parliament	7.750	−0.976	0.249	36
	[3.627]	(0.832)		
Share of hamlets represented in village parliament	0.843	0.054	0.339	36
	[0.202]	(0.056)		
Number of village parliament meetings in last year	5.714	−1.853	0.041**	44
	[4.689]	(0.878)		
Village parliament district system (1 = district, 0 = at large)	0.241	0.081	0.587	45
	[0.435]	(0.148)		
Number of previous KDP projects	1.875	−0.239	0.455	49
	[0.976]	(0.318)		

Notes: Column (1) presents the mean of the listed variable in the meeting villages, with standard deviations in brackets. Column (2) presents the difference between election and meeting villages, estimated with wave fixed effects, with robust standard errors in parentheses clustered at the village level. Column (3) shows the *p* value from a test of the null hypothesis that the listed variable is not different between elections and meeting villages. Column (4) shows the number of observations of the listed variable.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Estimation: using covariates additively

$$\hat{Y}_i^{obs} = \alpha + \tau \cdot W_i + \beta' \dot{X}_i + \epsilon_i$$

- $\dot{X}_i = X_i - \bar{X}$
- Using covariate deviations mean that the interpretation of τ and β does not change, just intercept α.

What is the issue with regression adjustment for experiments?

- If covariates don't correlate well with the outcome, or
- If there are too many covariates relative to sample size,
- Adjustment can result in noisier estimates, inconsistent SE estimation, finite sample bias.
- Also: fishing.

Addressing concerns

Lin (2013)'s proposed estimator

- Separate regressions of Y on X in both treatment and control.
- Use robust SEs.
- Using treatment $\times$ covariate interactions resolves risks to precision, even under model mispecification.
- Small sample bias—how much should we care about it?

Using covariates with full interactions

$$\hat{Y}_i^{obs} = \alpha + \tau \cdot W_i + \beta' \dot{X}_i + \gamma' \dot{X}_i \cdot W_i + \epsilon_i$$

- In general, γ don't have straightforward interpretation as differences in average treatment effects
- *Unless* all covariates are indicators and can be used to partition the population.

Interpretation with covariates

Using the Duflo et al. (2012) data,

$$\hat{Y}_i^{obs} = \underset{(0.02)}{0.59} + \underset{(0.03)}{0.192}\, W_i + \underset{(0.002)}{0.001}\, X_{1i} - \underset{(0.01)}{0.004}\, X_{2i} - \underset{(0.003)}{0.006}\, X_{1i} W_i + \underset{(0.01)}{0.017}\, X_{2i} W_i.$$

where X_1, X_2 are covariates.

Rules of thumb with covariates

- Pre-specify what you will measure and your main analyses.
- Don't include too many covariates relative to sample size.
- Checking covariate balance ex-post can be useful; but focus on using design decisions ex-ante to avoid great imblance on measured variables
- Use simpler forms of estimation: difference in means, stratified analysis, with robust SEs.

Stratified analysis

Using the Duflo et al. (2012) data,

$$\hat{Y}_i^{obs} = \underset{(0.02)}{0.60} + \underset{(0.03)}{0.188}\, W_i + \underset{(0.05)}{0.010}\, 1_{X_{1i}>37} - \underset{(0.06)}{0.06}\, 1_{X_{1i}>37} W_i.$$

where X_1 is teacher score.

Alternatives with machine learning methods

- Flexibly model response.
- Accounting for large numbers of covariates.
- Substitution estimators; estimation of heterogeneous treatment effects.
- Role of pre-registration when analysis decisions are data driven?

References I

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